

AF-4500 Series — Troubleshooting Guide

Companion to IOM-AF4500-REV-D

Document No: TSG-AF4500-REV-C | Issue Date: March 2026 | Atlas Fluid Systems B.V. — Global Technical Services

This guide lists common AF-4500 series symptoms, their probable causes, and the recommended corrective action. Always follow lockout-tagout procedure before any inspection or repair.

Symptom	Probable Cause	Corrective Action
Pump not delivering rated flow	Air leak in suction line; worn impeller	Check suction fittings and gaskets for air ingress; measure wear ring clearance
Excessive noise — "screaming"	Cavitation (insufficient NPSH available)	Reduce discharge pressure, foreign object in pump; replace bearings
High vibration at bearing housing	Coupling misalignment; unbalanced rotor	Re-check and correct coupling alignment; inspect/balance or replace impeller
Bearing housing overheating (>80°C)	Insufficient or excess lubricant; incorrect grease type	Re-grease per type, blocked cooling fins
Mechanical seal leakage	Worn seal faces; incorrect spring compression	Replace mechanical seal per SOP-114; verify seal installed to correct width
Motor trips on overload	Pump started against closed discharge valve	Start with discharge valve slightly open; monitor Section 5.2 of the IOM manual
Pump loses prime after startup	Air leak at suction gasket or seal; suspected excessive pump NPSH requirement	Inspect and replace pump NPSH gaskets; verify suction lift is within rated NPSH

Escalation

If a symptom persists after the corrective action above, or is not listed here, log a ticket in Atlas FieldOps and escalate to Technical Support: Rotterdam desk (+31 10 555 0142, 08:00–18:00 CET) or Houston desk (+1 713 555 0198, 24/7 for critical customer downtime). Reference the pump serial number and this document number when escalating.

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